Quantitative Evaluation of Factory-to-Customer Shipping Route Efficiency and Logistics Bottlenecks in Wholesale Confectionery Distribution

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Abstract

This study evaluates the end-to-end supply chain efficiency of Nassau Candy Distributor across multi-factory shipping networks. By calculating route-level lead times, delay frequencies, and normalized efficiency scores across 5 production facilities and nationwide delivery routes, this paper identifies critical shipping bottlenecks, analyzes mode-specific cost-versus-speed trade-offs, and provides strategic recommendations to optimize fulfillment performance.

1. Introduction & Methodology

To analyze logistics performance across regional networks, order transactions were paired with product-factory sourcing mappings.

- **Shipping Lead Time Calculation:** Calculated as $\text{Shipping Lead Time} = \text{Ship Date} - \text{Order Date}$ (in calendar days).
- **Normalized Route Efficiency Score:** Normalized on a scale of 0–100 using Min-Max scaling based on route average lead times:

$$\text{Efficiency Score} = 100 \times \left(1 - \frac{\text{Lead Time} - \min(\text{Lead Time})}{\max(\text{Lead Time}) - \min(\text{Lead Time})}\right)$$

- **Delay Threshold:** Orders exceeding a 4-day shipping duration were flagged as delayed to determine regional delay frequencies.

2. Key Analytical Findings

1. **Factory Lead Time Variances:** Direct origin-to-destination mapping reveals that longer transit times correlate strongly with multi-state long-haul routes originating from centralized hubs (e.g., *Sugar Shack* and *Secret Factory*).
2. **Ship Mode Trade-Offs:** Standard Class shipping exhibits high variance in lead time (3 to 7 days), whereas Expedited modes reliably achieve delivery within 1 to 2 days, highlighting a clear margin-versus-speed tradeoff.
3. **Regional Bottlenecks:** Destination states located farthest from primary manufacturing clusters experience up to 40% higher delay frequencies during peak fulfillment cycles.

3. Strategic Supply Chain Recommendations

- **Regional Safety Stock Allocation**: Reallocate high-demand inventory closer to lagging regional customer clusters to reduce cross-country transit steps.
- **Dynamic Route Optimization**: Shift non-urgent orders from bottlenecked highways to intermodal rail freight to reduce logistics expenditures while maintaining target delivery SLAs.